

KNOWLEDGE SYSTEMS WORKING GROUP

Enabling Semantic Interoperability

Across Engineering Disciplines

Through vocabularies, taxonomies, and ontologies that humans can understand and machines can process

THE PROBLEM

50+ Working Groups. One Language Barrier.



Incompatible Terminologies

"Requirement" means different things across Requirements WG, MBSE Initiative, and Healthcare WG



Redundant Development

Multiple WGs independently build overlapping frameworks without coordination



Cross-Domain Blind Spots

No infrastructure exists to synthesize expertise across disciplines



AI Systems Lack Grounding

Without semantic structure, AI tools cannot reliably interpret SE knowledge

The Cost of Fragmentation

Lost Innovation

Critical insights emerge at domain boundaries—but without shared semantics, these connections remain invisible

Redundant Effort

Each WG reinvents conceptual frameworks that could be shared, wasting collective expertise

Diminished Impact

INCOSE's collective knowledge is worth more than the sum of isolated parts—if we can connect it

The primary benefit is human comprehension of human knowledge.

AI enablement is a secondary benefit that follows from good semantic architecture.

Knowledge Engineering for INCOSE

What We Build



Controlled Vocabularies

Standardized terms with agreed definitions across all Working Groups



Taxonomies

Hierarchical classification systems that organize SE concepts



Ontologies

Formal semantic relationships enabling machine-readable knowledge

How We Work

- Evangelize semantic approaches with real demos and case studies
- Facilitate ontology development across INCOSE Working Groups
- Identify shared conceptual patterns while preserving domain context
- Provide progressive integration pathways through high-value use cases

A Secondary Benefit, Done Right

Good semantic architecture naturally enables AI systems.

We're not building for AI—we're building knowledge structures that AI can leverage.

What Semantic Structure Enables

- ✓ Contextual grounding for AI queries about SE domains
- ✓ Retrieval-augmented generation with curated, verified knowledge
- ✓ Ontological constraints that improve AI output reliability

What We're NOT Claiming

- × Production-ready neurosymbolic AI systems (notional architecture)
- × Specific ROI figures (benefits are real but context-dependent)
- × That "prompt engineering" is a durable discipline

*Collaboration with AI WG is critical to developing formalized, trustworthy perspectives
on AI integration that go beyond what general sources provide.*

THE BENEFITS

What Your Working Group Gains



Shared Understanding

Common conceptual models let Working Groups understand each other's domains without manual translation



Coordination Infrastructure

Semantic frameworks provide the foundation for cross-group initiatives and joint deliverables



Unambiguous Communication

Controlled vocabularies eliminate the "same word, different meaning" problem across disciplines



Collective Impact

INCOSE's expertise becomes more valuable when it can be connected and synthesized

THE ASK

Join Us in Building This Together

1

Participate

Share your WG's key terms, concepts, and pain points around cross-domain communication

2

Collaborate

Join cross-WG sessions to map semantic connections and identify shared patterns

3

Champion

Be a liaison between KSWG and your Working Group to drive adoption

This work requires patience—building interoperable ontologies across 50+ Working Groups is a multi-year effort. But every journey starts with willing collaborators.

THANK YOU

Let's talk after the session

Knowledge Systems Working Group

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